Assignment 5: Data Visualization

Tallulah Bowden

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
#load packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at /home/guest/Environmental Data Exploration/EDE_Fall2025
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
library(ggplot2)
library(ggthemes)
```

```
##
## Attaching package: 'ggthemes'
##
## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```r
#check WD
#getwd()
#here()

#read in datasets
Peter_Paul <- read.csv(
"Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv",
  stringsAsFactors = T)
NEON <- read.csv(
  "Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv",
  stringsAsFactors = T)
#2
#check date classes
class(NEON$collectDate)
```

```
## [1] "factor"
```

```r
class(Peter_Paul$sampledate)
```

```
## [1] "factor"
```

```r
#change from factor to date
NEON$collectDate <- ymd(NEON$collectDate)
Peter_Paul$sampledate <- ymd(Peter_Paul$sampledate)

#check date classes again
class(NEON$collectDate)
```

```
## [1] "Date"
```

```
class(Peter_Paul$sampledate)
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
my_theme <- theme_base()  +
  theme(
    line = element_line(
      color = "black",
      linewidth = 1),
    legend.background = element_rect(
      color = "grey",
      fill = "light blue"),
    lenged.title = element_text(
      color = "blue"),
    axis.title.x = element_text(
      color = "black"),
    axis.title.y = element_text(
      color = "black"),
    legend.position = "right",
    complete = T)

theme_set(my_theme)
```

Create graphs

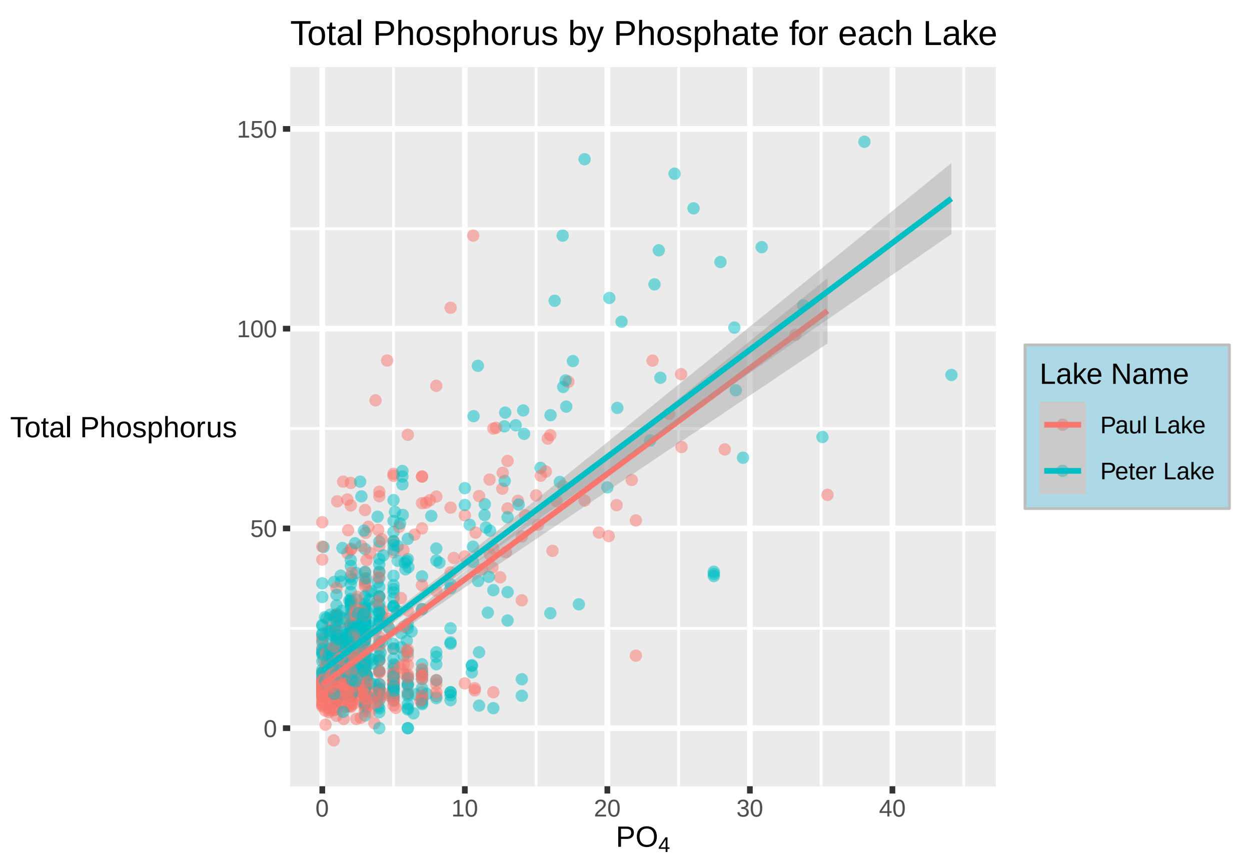
For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
phosphorus_phosphate <- Peter_Paul %>%
ggplot(
  mapping =
    aes(x = po4,
```

```
         y = tp_ug,
         color = lakename))+
  geom_point(alpha = 0.5)+
  geom_smooth(method = lm)+
  xlim(0,45)+
  labs(x = expression("PO" [4])) +
  labs(color = "Lake Name")+
  labs(y = "Total Phosphorus")+
  labs(title = "Total Phosphorus by Phosphate for each Lake")

phosphorus_phosphate
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 21947 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning in plot_theme(plot): The `lenged.title` theme element is not defined in
## the element hierarchy.

## Warning: Removed 21947 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



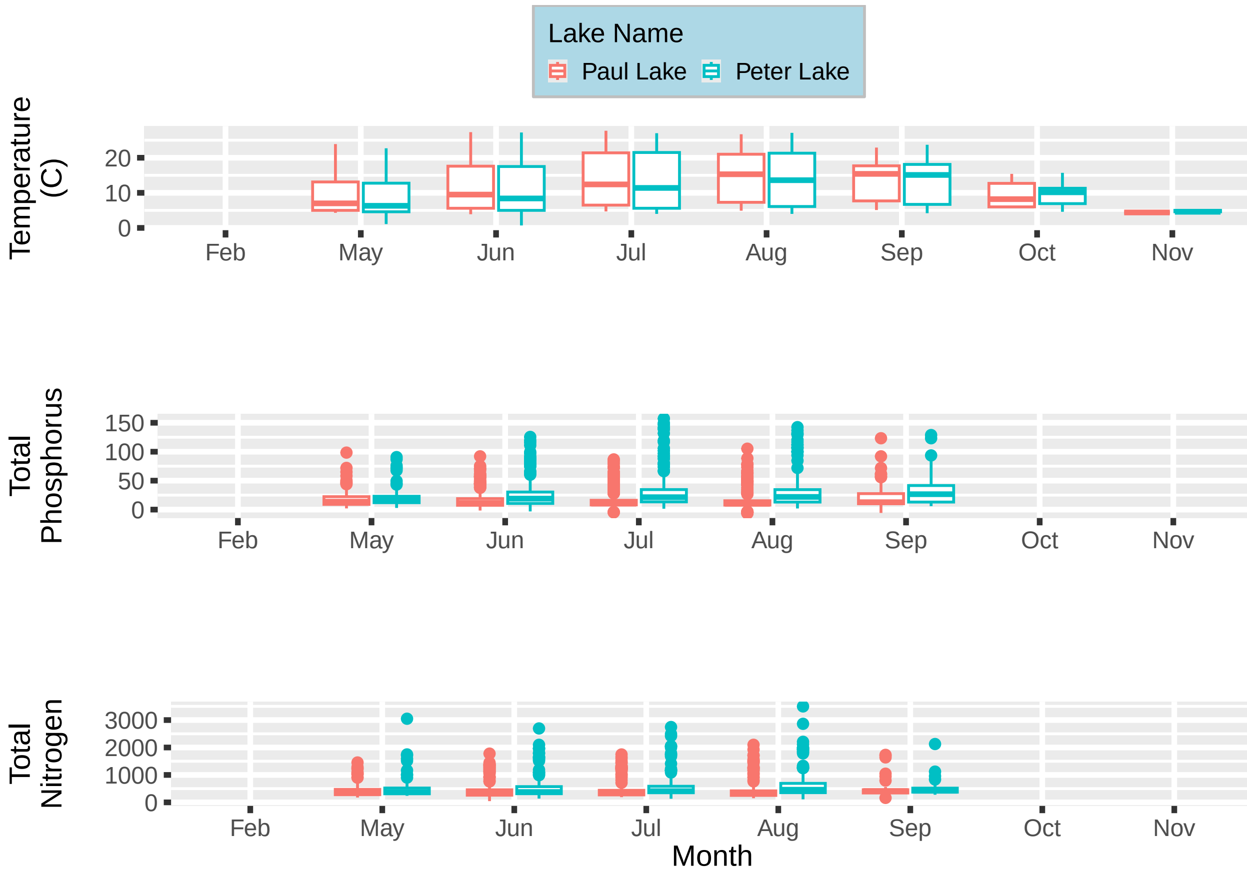
5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
Peter_Paul_month <-
  mutate(Peter_Paul,
    month_num = month(sampledate),
    month_name_abb = month(sampledate, label = TRUE),
    month_name_full = month(sampledate, label = TRUE, abbr = FALSE)
  )

temp_plot <- Peter_Paul_month %>%
  ggplot(
    mapping =
      aes(x = month_name_abb,
          y = temperature_C,
          color = lakename))+
  geom_boxplot()+
  labs(x = "Month") +
labs(y = "Temperature\n(C)") +
labs(color = "Lake Name") +
  theme(legend.position = "top",
        axis.title.x = element_blank(),
        legend.key.size = unit(0.5, "lines"),
        legend.title = element_text(size = 10),
        legend.title.position = "top",
        axis.title.y = element_text(angle = 90,
                                    margin =
                                      margin(r = 15)),
        plot.margin = unit(c(0, 0, 0, 0), "cm"))



#temp_plot

TP_plot <- Peter_Paul_month %>%
  ggplot(
    mapping =
      aes(x = month_name_abb,
          y = tp_ug,
          color = lakename))+
  geom_boxplot()+
  labs(x = "Month")+
  labs(y = "Total\nPhosphorus")+
  labs(color = "Lake Name")+
  theme(axis.title.x = element_blank(),
        legend.position = "none",
        axis.title.y = element_text(angle = 90,
```

```
                                    margin =
                                      margin(r = 15)),
        plot.margin = unit(c(0, 0, 0, 0), "cm"))

#TP_plot

TN_plot <- Peter_Paul_month %>%
  ggplot(
    mapping =
      aes(x = month_name_abb,
          y = tn_ug,
          color = lakename))+
  geom_boxplot()+
  labs(x = "Month")+
  labs(y = "Total\nNitrogen")+
  labs(color = "Lake Name")+
  theme(legend.position = "none",
        axis.title.y = element_text(angle = 90,
                                    margin =
                                      margin(r = 15)),
        plot.margin = unit(c(0, 0, 0, 0), "cm"))

#TN_plot

plot_grid(
  temp_plot,
  TP_plot,
  TN_plot,
  nrow = 3,
  align = 'h')
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning in plot_theme(plot): The `lenged.title` theme element is not defined in
## the element hierarchy.

## Warning: Removed 20729 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning in plot_theme(plot): The `lenged.title` theme element is not defined in
## the element hierarchy.

## Warning: Removed 21583 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning in plot_theme(plot): The `lenged.title` theme element is not defined in the element hierarchy
## The `lenged.title` theme element is not defined in the element hierarchy.
```



Question: What do you observe about the variables of interest over seasons and between lakes?

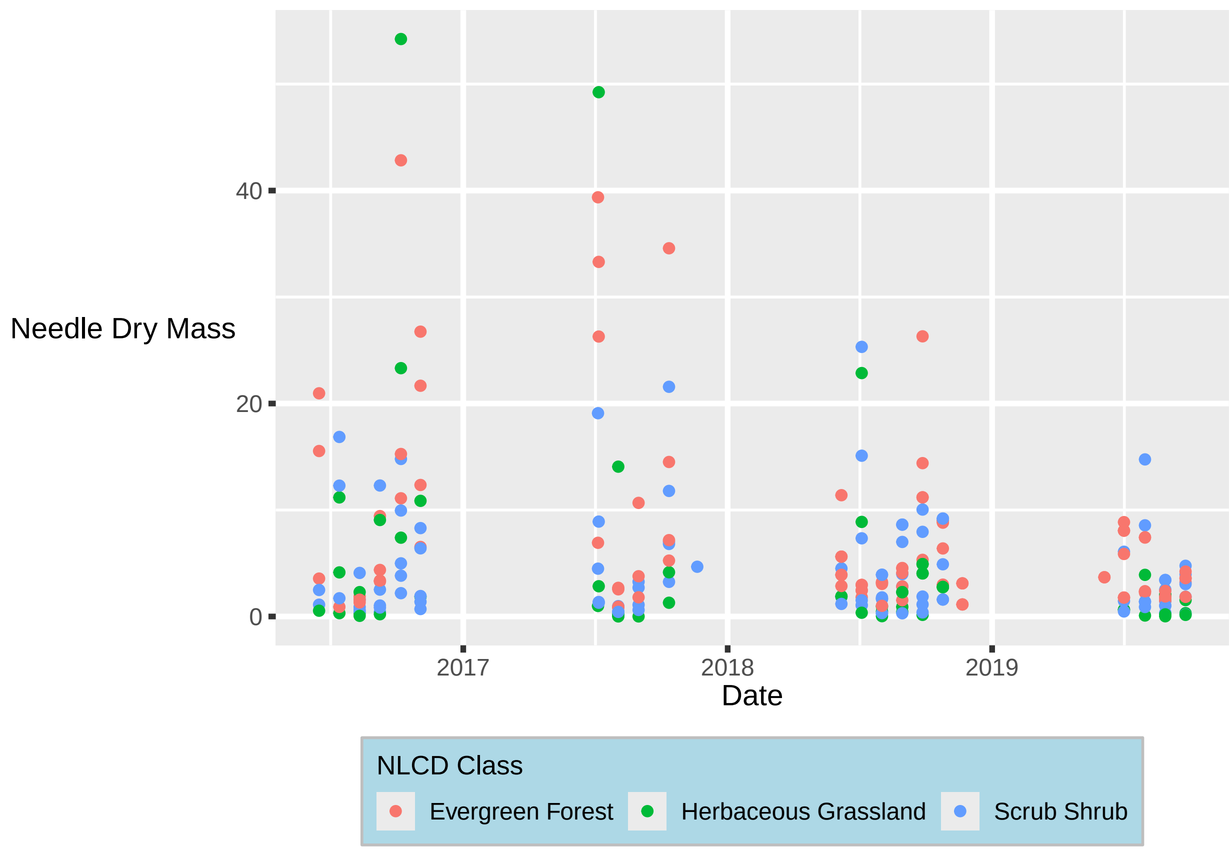
> Answer: Temperature follows a curve with May and June having colder temperatures and July through September having hotter temperatures. Going into October, the temperature begins to drop again. Paul lake seems to have slightly higher temperatures at nearly all months except October and November Total phosphorus also starts with lower values in May and increases very slightly through September The interquartile ranges also increase over this time.Paul lake consistently has lower total phosphorus values. Total nitrogen follows a similar trend to total phosphorus. Paul lake also has lower values than Peter lake.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
needle_plot <- NEON %>%
  filter(functionalGroup == "Needles")%>%
  ggplot(mapping =
           aes(x = collectDate,
               y = dryMass,
               color = nlcdClass))+
  geom_point()+
  labs(x = "Date")+
```

```
  labs(y = "Needle Dry Mass")+
  labs(color = "NLCD Class")+
  scale_color_discrete(
    labels = c("Evergreen Forest", "Herbaceous Grassland", "Scrub Shrub")
  )+
  theme(legend.position = "bottom",
        legend.key.size = unit(1, "lines"),
        legend.title = element_text(size = 10),
        legend.title.position = "top")

needle_plot
```

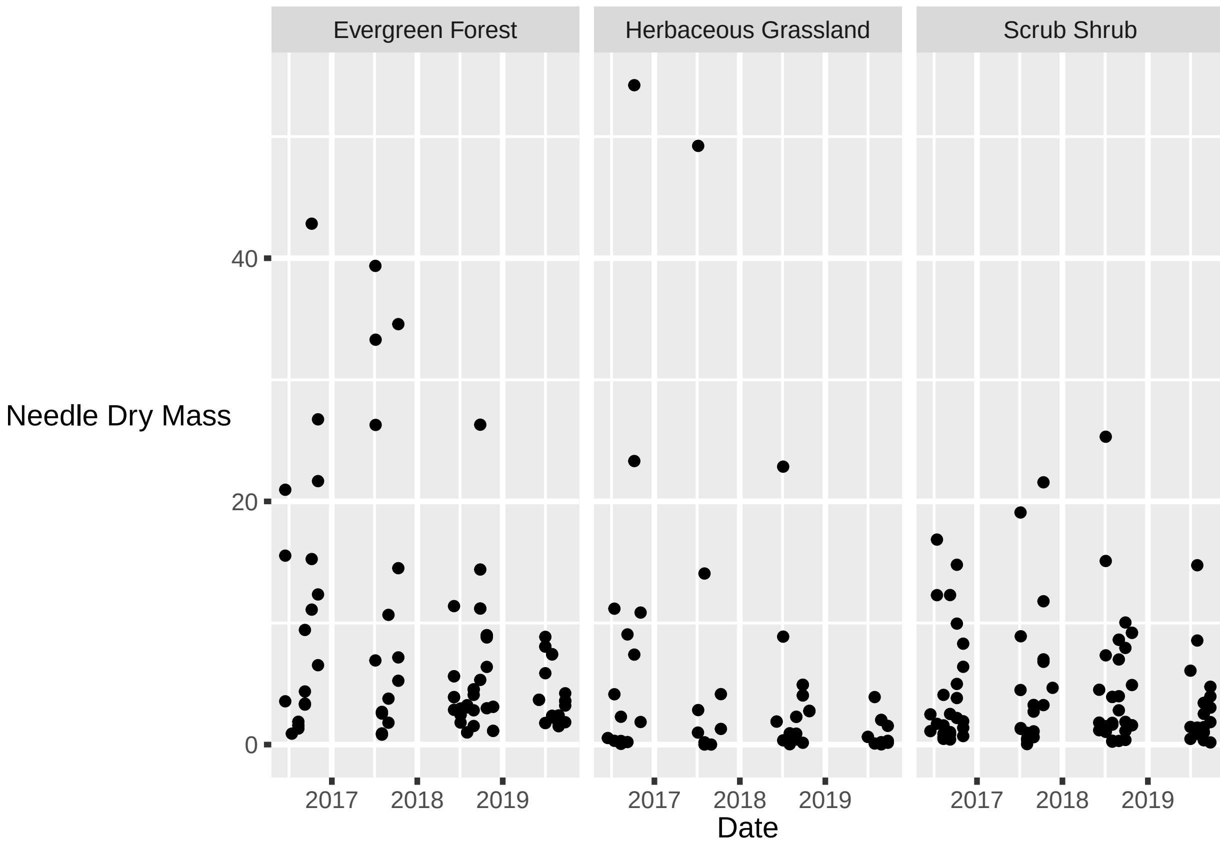
```
## Warning in plot_theme(plot): The ‘lenged.title‘ theme element is not defined in
## the element hierarchy.
```




```
#7
needle_faceted_plot <- NEON %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(
    mapping =
      aes(x = collectDate,
          y = dryMass))+
  geom_point()+
  facet_wrap(vars(nlcdClass),
```

```
            ncol = 3,
            labeller = labeller(nlcdClass =
                         c("evergreenForest" = "Evergreen Forest",
                         "grasslandHerbaceous" = "Herbaceous Grassland",
                         "shrubScrub" = "Scrub Shrub")))+
  labs(x = "Date")+
  labs(y = "Needle Dry Mass")

needle_faceted_plot
```

```
## Warning in plot_theme(plot): The 'lenged.title' theme element is not defined in
## the element hierarchy.
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer: I think the faceted plot (7) is more effective in this analysis. Although I like having plots with color, in the first plot it was hard to have to go to the legend and find which color represented which NLCD class and then go back to the plot and find which date it represented and then what dry mass it was associated with. There were a lot of steps to get meaningful information out of that plot and it would be even harder to look at trends. In the faceted plot, it is much easier to look at one of the facets and be able to tell what is happening with needle dry mass over our collection time.